



SMT Power Inductors – MSS1260



- 12.3 × 12.3 mm footprint; 6 mm high shielded inductors
- Low DCR and excellent current handling

Designer's Kit C360 contains 3 each of all values.

Core material Ferrite

Core and winding loss See www.coilcraft.com/coreloss

Terminations RoHS compliant matte tin over nickel over phos bronze. Other terminations available at additional cost.

Weight: 2.8 – 3.3 g

Ambient temperature –40°C to +85°C with I_{rms} current, +85°C to +125°C with derated current

Storage temperature Component: –40°C to +125°C.

Packaging: –40°C to +80°C

Resistance to soldering heat Max three 40 second reflows at +260°C, parts cooled to room temperature between cycles

Moisture Sensitivity Level (MSL) 1 (unlimited floor life at <30°C / 85% relative humidity)

Failures in Time (FIT) / Mean Time Between Failures (MTBF)

38 per billion hours / 26,315,789 hours, calculated per Telcordia SR-332

Packaging 500/13" reel; Plastic tape: 24 mm wide, 0.35 mm thick, 16 mm pocket spacing, 6.3 mm pocket depth

PCB washing Only pure water or alcohol recommended

Part number ¹	Inductance ² (μ H)	DCR ³ (mOhms)		SRF typ ⁴ (MHz)	Isat (A) ⁵			Irms (A) ⁶	
		typ	max		10% drop	20% drop	30% drop	20°C rise	40°C rise
MSS1260-102NL	1.0 $\pm$ 30%	5.8	6.5	100	19.1	21.4	22.7	6.00	8.00
MSS1260-152NL	1.5 $\pm$ 30%	8.8	9.8	80.0	15.0	16.6	17.6	6.00	7.50
MSS1260-222NL	2.2 $\pm$ 30%	11.5	12.8	55.0	11.7	13.0	13.9	5.50	7.00
MSS1260-332NL	3.3 $\pm$ 30%	12.6	14.0	42.0	10.4	11.7	12.5	5.00	7.00
MSS1260-472ML	4.7 $\pm$ 20%	13.9	15.5	33.0	9.22	10.1	10.8	4.50	7.00
MSS1260-562ML	5.6 $\pm$ 20%	14.9	16.6	30.0	7.86	9.02	9.74	4.00	6.40
MSS1260-682ML	6.8 $\pm$ 20%	16.6	18.5	27.0	7.40	8.26	8.80	3.80	5.90
MSS1260-822ML	8.2 $\pm$ 20%	20.2	22.5	26.0	7.10	7.96	8.50	3.40	4.80
MSS1260-103ML	10 $\pm$ 20%	21.5	23.9	22.0	6.18	6.92	7.40	3.00	4.00
MSS1260-123ML	12 $\pm$ 20%	24.5	27.3	20.0	5.18	5.94	6.42	2.80	3.70
MSS1260-153ML	15 $\pm$ 20%	30.7	34.2	18.0	4.80	5.40	5.78	2.60	3.50
MSS1260-183ML	18 $\pm$ 20%	35.4	39.4	16.0	4.58	5.22	5.62	2.50	3.30
MSS1260-223ML	22 $\pm$ 20%	36.6	40.7	15.0	4.06	4.64	4.96	2.30	3.10
MSS1260-273ML	27 $\pm$ 20%	51.3	57.0	13.0	3.52	3.96	4.28	2.10	2.90
MSS1260-333ML	33 $\pm$ 20%	54.9	61.0	12.4	3.22	3.74	4.02	2.00	2.70
MSS1260-393ML	39 $\pm$ 20%	58.0	64.5	12.0	3.08	3.56	3.80	1.90	2.60
MSS1260-473ML	47 $\pm$ 20%	80.1	89.0	11.6	2.66	3.04	3.30	1.85	2.50
MSS1260-563ML	56 $\pm$ 20%	82.5	91.7	10.5	2.54	2.96	3.14	1.75	2.40
MSS1260-683ML	68 $\pm$ 20%	94.5	105.0	10.0	2.40	2.70	2.94	1.70	2.30
MSS1260-823ML	82 $\pm$ 20%	131.6	146.3	8.6	2.16	2.46	2.64	1.60	2.20
MSS1260-104ML	100 $\pm$ 20%	141.8	157.6	7.8	1.88	2.16	2.32	1.50	2.10
MSS1260-124KL	120 $\pm$ 10%	193.3	214.8	6.8	1.70	1.92	2.10	1.38	1.85
MSS1260-154KL	150 $\pm$ 10%	215.4	239.4	6.4	1.58	1.80	1.98	1.20	1.66
MSS1260-184KL	180 $\pm$ 10%	254.2	282.5	6.1	1.40	1.60	1.72	1.14	1.58
MSS1260-224KL	220 $\pm$ 10%	314.1	349.0	5.5	1.28	1.44	1.56	1.00	1.42
MSS1260-274KL	270 $\pm$ 10%	368.8	409.8	4.3	1.10	1.26	1.38	0.90	1.45
MSS1260-334KL	330 $\pm$ 10%	481.3	534.8	4.0	1.00	1.14	1.24	0.84	1.16
MSS1260-394KL	390 $\pm$ 10%	517.5	575.0	3.6	0.93	1.06	1.15	0.78	1.08
MSS1260-474KL	470 $\pm$ 10%	721.2	801.4	3.0	0.87	0.99	1.06	0.70	0.96
MSS1260-564KL	560 $\pm$ 10%	773.1	859.0	2.8	0.81	0.92	1.00	0.64	0.88
MSS1260-684KL	680 $\pm$ 10%	867.6	964.0	2.6	0.74	0.85	0.92	0.58	0.80
MSS1260-824KL	820 $\pm$ 10%	1158	1287	2.5	0.66	0.76	0.81	0.53	0.73
MSS1260-105KL	1000 $\pm$ 10%	1273	1415	2.4	0.60	0.69	0.74	0.48	0.68

- Please specify **termination** and **packaging** codes:

MSS1260-184K L D

Termination: L = RoHS compliant matte tin over nickel over phos bronze.

Special order:

T = RoHS tin-silver-copper (95.5/4/0.5) or S = non-RoHS tin-lead (63/37).

Packaging: D = 13" machine-ready reel. EIA-481 embossed plastic tape (500 parts per full reel).

B = Less than full reel. In tape, but not machine ready. To have a leader and trailer added (\$25 charge), use code letter D instead.

- Inductance tested at 100 kHz, 0.1 Vrms, 0 Adc using an Agilent/HP 4263B LCR meter or equivalent.
 - DCR measured on a micro-ohmmeter and a Coilcraft CCF858 test fixture.
 - SRF measured using Agilent/HP 4191A or equivalent.
 - DC current at which the inductance drops the specified amount from its value without current.
 - Current that causes the specified temperature rise from 25°C ambient.
 - Electrical specifications at 25°C.
- Refer to Doc 362 "Soldering Surface Mount Components" before soldering.

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Please check our website for latest information.

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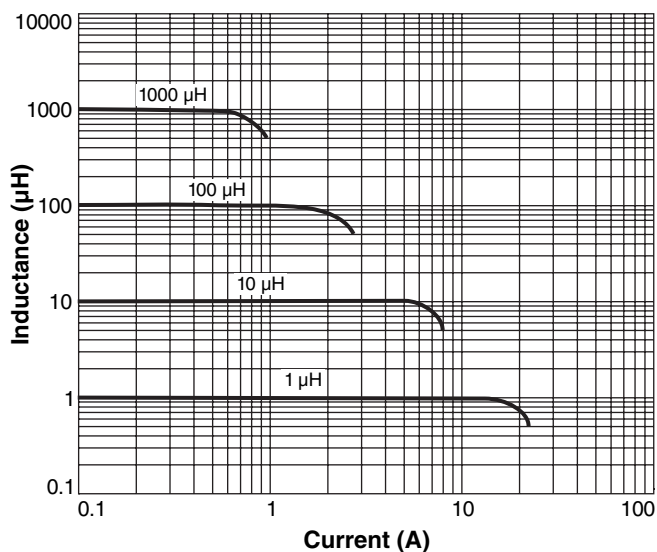
1102 Silver Lake Road Cary, Illinois 60013 Phone 847/639-6400 Fax 847/639-1469

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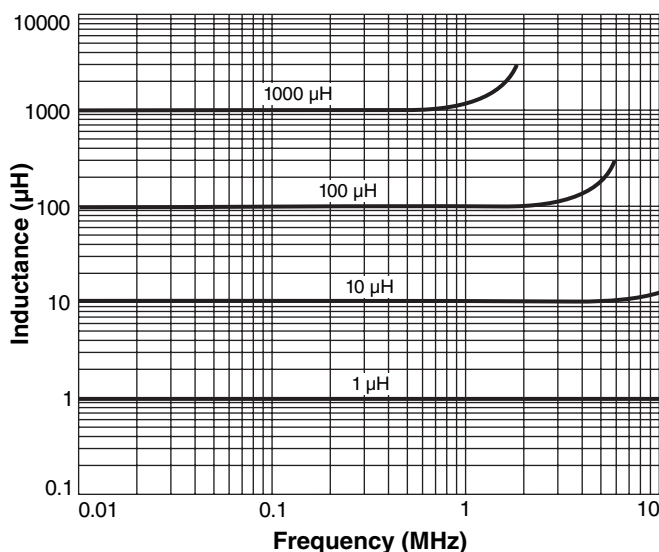


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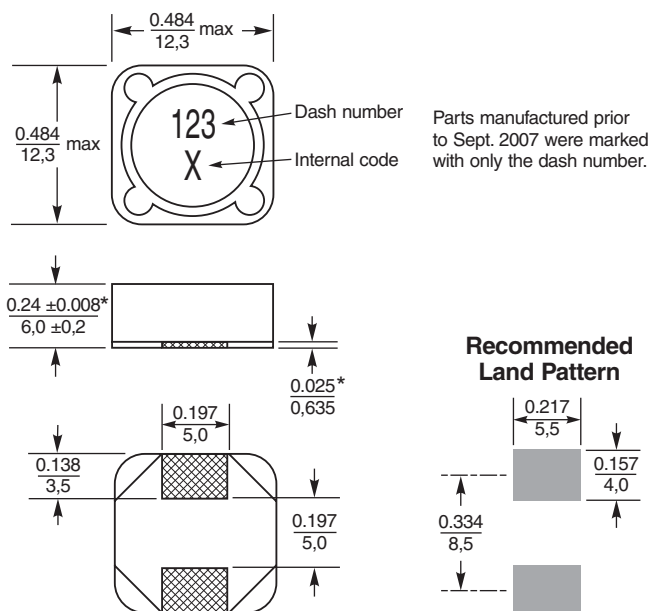
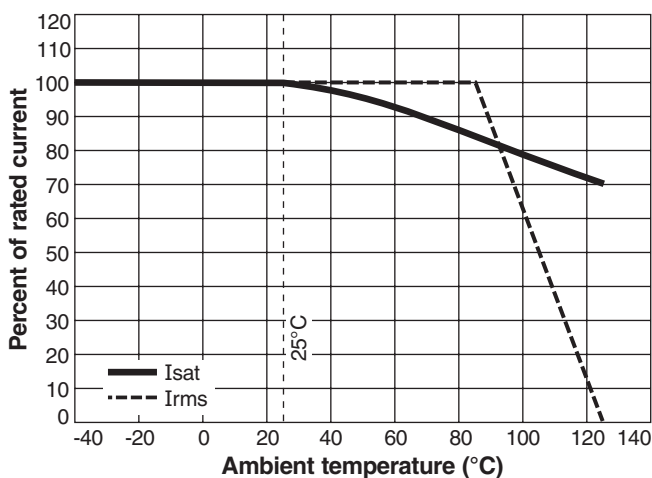
Typical L vs Current



Typical L vs Frequency



Typical Current Derating



* For optional tin-lead and tin-silver-copper terminations, dimensions are for the mounted part. Dimensions before mounting can be an additional 0.012 inch (0.3 mm).

Dimensions are in $\frac{\text{inches}}{\text{mm}}$

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Specifications subject to change without notice.
Please check our website for latest information.

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